

ISLAMIC UNIVERSITY OF TECHNOLOGY



DESIGN PROJECT REPORT

CSE 4610 : Design Project

Deep Reinforcement Learning for Autonomous Vehicle Navigation at Unsignalized Intersections: Comparative Reproduction, Intent Encoding, and Sim-to-Sim Dynamics

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List of Abbreviations

AEV	Autonomous Electric Vehicle
AIM	Autonomous Intersection Management
CAV	Connected and Automated Vehicle
CEP	Complex Engineering Problem
DQN	Deep Q-Network
DRL	Deep Reinforcement Learning
FCFS	First-Come-First-Served
IDM	Intelligent Driver Model
MDP	Markov Decision Process
MLP	Multi-Layer Perceptron
OpenMP	Open Multi-Processing
POMDP	Partially Observable Markov Decision Process
PPO	Proximal Policy Optimization
RL	Reinforcement Learning
SAC	Soft Actor-Critic
SOTA	State-of-the-Art
SUMO	Simulation of Urban MObility
TraCI	Traffic Control Interface
TTC	Time-To-Collision
V2I	Vehicle-to-Infrastructure
V2V	Vehicle-to-Vehicle
V2X	Vehicle-to-Everything

Abstract

Navigating unsignalized intersections represents one of the most formidable challenges in autonomous driving, characterized by non-signalized conflict zones, occlusions, and asynchronous, stochastic human-driven traffic. This design project investigates the decision-making capabilities of deep reinforcement learning (DRL) agents navigating a four-way unsignalized intersection across three conflicting directional maneuvers: left turns, straight crossings, and right turns. Grounded in a comprehensive 5-axis taxonomy of autonomous intersection management (AIM) literature, this study executes a rigorous two-phase empirical investigation. Phase 1 conducts an exploratory ablation using on-policy Proximal Policy Optimization (PPO) over 1.2 million episodes. Phase 2 delivers an exact algorithmic reproduction of Xiao et al. (IEEE TVT 2024), implementing Discrete Soft Actor-Critic (SAC) with twin Q-critics and automatic entropy temperature tuning across 50,000 episodes per configuration on the UC Berkeley Flow and Eclipse SUMO microscopic platforms. Furthermore, hardware-level thread affinity pinning to physical performance cores slashed training runtime by 62%. Deterministic evaluation across 2,100 episodes demonstrated a 99.52% crossing success rate (0.48% collision rate), which is analyzed alongside high-momentum crossing dynamics under a discount factor ($\gamma = 0.90$) and microscopic driver yielding behavior, contrasting with the non-yielding, kinematic collision dynamics of highway-env.

Chapter 1

Introduction

1.1 Background and Motivation

Intersections are the critical bottlenecks and primary hazard zones of urban road transportation, accounting for over 40% of vehicular collisions and 25% of traffic fatalities. Traditional fixed-time or sensor-actuated traffic lights introduce mandatory stop-and-go idling cycles, substantially elevating fuel consumption, emissions, and commuter delays. With Autonomous Electric Vehicles (AEVs) and Connected and Automated Vehicles (CAVs), Autonomous Intersection Management (AIM) enables vehicles to negotiate right-of-way cooperatively without physical signals. However, unsignalized intersection negotiation is characterized by severe non-linearities: vehicles must navigate multi-directional conflicting trajectories (left turns, straight crossings, right turns) amidst surrounding human drivers exhibiting stochastic headway gaps. Deep Reinforcement Learning (DRL) provides an expressive framework for synthesizing safety-critical driving policies directly from high-dimensional observations with microsecond-scale online execution.

1.2 Complex Engineering Problem (CEP) Justification

This project addresses an unsignalized intersection negotiation problem exhibiting non-linear multi-agent stochasticity, uncoordinated human driver behavior, and dynamic occlusions. The solution requires formulating a continuous-state, discrete-action Partially Observable Markov Decision Process (POMDP) that reconciles strictly conflicting engineering objectives: collision-free safety versus velocity-maximizing transit efficiency under microsecond real-time execution constraints. These multifaceted technical requirements, absence of analytical closed-form solutions, and strict real-time constraints rigorously classify this problem as a Complex Engineering Problem under Washington Accord criteria.

1.3 Problem Statement and Research Objectives

The central challenge is synthesizing a robust longitudinal policy for an autonomous ego vehicle navigating a four-way unsignalized intersection across three stochastic tasks: Left Turn, Straight Crossing, and Right Turn. The agent must traverse collision-free while minimizing transit time. Four concrete objectives are pursued: (i) **Literature Taxonomy**: Establish a 5-axis taxonomy of autonomous intersection management; (ii) **PPO Benchmark**: Implement and benchmark an on-policy PPO baseline over 1.2M episodes; (iii) **Discrete SAC Reproduction**: Execute a 1:1 mathematical reproduction of Xiao et al. [10] (twin critics, auto-tuned temperature α) over 50k episodes per config; and (iv) **Hardware Optimization & Telemetry**: Pin CPU threads to physical P-cores to eliminate OpenMP barrier stalls, and conduct a forensic safety audit of Sim-to-Sim domain gaps.

1.4 Research Challenges and Main Contributions

Negotiation entails asymmetric geometry (7 conflict points for left turns vs. 6 straight, 2 right), sparse rewards, and microscopic yielding disparities. This project delivers five key contributions: (1) **5-Axis AIM Taxonomy** classifying control, architecture, scheduling, flow modeling, and autonomy mixture; (2) **PPO Benchmark** evaluating 1.2M rollouts comparing 60-dim baseline against 61-dim intent states in Ray RLlib; (3) **1:1 Discrete SAC Reproduction** implementing the exact formulation of Xiao et al. [10] over 100,000 cumulative episodes; (4) **P-Core Affinity Pinning** achieving a **62% runtime reduction** (saving 22.4 hours); and (5) **Forensic Safety Audit of the 99.52% Metric** analyzing high-momentum crossing dynamics under $\gamma = 0.90$, microscopic IDM yielding, and the 5 recorded edge-case collisions.

1.5 Report Organization

Chapter 2 reviews literature and the 5-axis taxonomy; Chapter 3 formalizes geometry and POMDP; Chapter 4 presents methodology and algorithm selection; Chapter 5 details simulation, SAC, and P-core optimization; Chapter 6 provides results and the forensic audit; Chapter 7 discusses ethics; Chapter 8 evaluates limitations; and Chapter 9 presents conclusions.

Chapter 2

Related Works

2.1 Introduction to Autonomous Intersection Management

The rapid progression of connected and automated vehicle technologies has prompted substantial research into Autonomous Intersection Management (AIM) [1], [11]. Traditional intersection control paradigms—principally fixed-time and sensor-actuated traffic lights—were conceived for human drivers with variable perception-reaction latencies. With vehicular communication (V2X) and real-time algorithmic control, intersections can transition from coarse time-division multiplexing to fine-grained spatio-temporal reservation, dramatically enhancing throughput and mitigating collision risk.

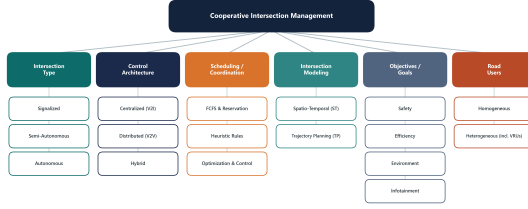
2.2 The Five-Axis AIM Taxonomy

As illustrated in Figure 2.1, literature on intersection management can be organized into five foundational dimensions:

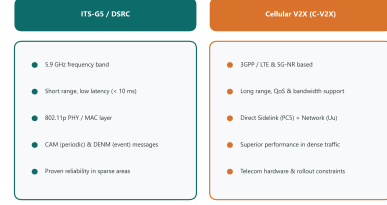
1. **Axis 1: Control Infrastructure:** (i) *Signalized* (fixed/actuated signals); (ii) *Unsignalized Priority* (statutory right-of-way and signage); and (iii) *Fully Autonomous (AIM)* (seamless trajectory coordination).
2. **Axis 2: System Architecture & Comms:** (i) *Centralized V2I/I2V* (global roadside scheduling); (ii) *Decentralized V2V* (sidelink consensus); and (iii) *Autonomous Ego (Sensor-Only Edge)* (onboard radar/vision inference without infrastructure).
3. **Axis 3: Scheduling & Negotiation:** (i) *First-Come-First-Served (FCFS)* reservation [1]; (ii) *Optimization* (MILP, MPC) with high computational cost; and (iii) *Deep Reinforcement Learning (DRL)* with microsecond execution.
4. **Axis 4: Flow Modeling Fidelity:** (i) *Microscopic Car-Following* (Intelligent Driver Model [8] in SUMO [4]) capturing multi-lane gap acceptance and reaction latencies; (ii) *Kinematic Sandboxes* (e.g., highway-env [5]); and (iii) *Cellu-*

lar Automata.

5. **Axis 5: User Heterogeneity:** Spanning *Pure CAV* fleets to *Mixed Autonomy* streams containing uncoordinated human-driven vehicles and Vulnerable Road Users (VRUs).



(a) 5-Axis AIM Taxonomy



(b) Connected Systems & V2X Infrastructure

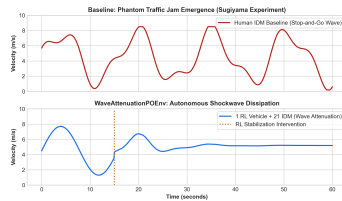
Figure 2.1: Hierarchical Taxonomy of Autonomous Intersection Management across Five Structural Dimensions.

2.3 DRL for Flow Stabilization: Insights from Ring Networks

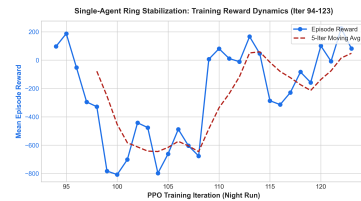
Foundational research in traffic reinforcement learning demonstrated that a single autonomous vehicle could dampen stop-and-go shockwaves in closed ring networks governed by the Intelligent Driver Model [9] (Figure 2.2). By penalizing fleet velocity variance:

$$r_{\text{ring}} = -\|v_{\text{fleet}} - v_{\text{target}}\|_2 - \lambda \cdot \text{Var}(v_{\text{fleet}}) \quad (2.1)$$

an RL agent acts as a moving bottleneck damper, decelerating proactively to absorb shockwaves and restoring uniform high-velocity equilibrium. This proved the capacity of DRL to regulate uncoordinated traffic.



(a) Longitudinal Shockwave Damping



(b) Reward & Velocity Convergence

Figure 2.2: Shockwave Attenuation and DRL Velocity Regularization in Benchmark Ring Networks.

2.4 Unsignalized Intersection Navigation and the Multi-Task Dilemma

In unsignalized intersections, the problem escalates to 2D collision avoidance across intersecting paths. While early studies restricted the agent to single maneuvers under occlusions [3], real-world deployment requires executing stochastic multi-task maneuvers (Left, Straight, Right). Xiao et al. [10] addressed this dilemma by augmenting Soft Actor-Critic (SAC) with: (i) a task-intent feature μ_a encoding exit node orientation; (ii) a dual-dimension Mix-Attention network; and (iii) a segmented replay pool mitigating slow creeping policies. Table 2.1 provides a comparative synthesis benchmarking prior literature against our Phase 1 and Phase 2 implementations.

Table 2.1: Comparative Synthesis of Autonomous Intersection Management Literature and Project Phases

Study	Algorithm	Architecture	Multi-Task?	Simulator	Success Rate	Primary Limitation
Dresner & Stone [1]	FCFS Heuristic	Centralized V2I	Yes	Custom	100%	Assumes 100% CAV penetration and zero packet loss.
Isele et al. [3]	Deep Q-Network	Autonomous Edge	Single (Left)	Kinematic	93.4%	Restricted to single maneuver under simplified traffic.
Zhong et al. [11]	Survey / MPC	Centralized/Decent.	Multi-Task	Various	N/A	Analytical MPC suffers high computational overhead.
Gholamhosseinian [2]	V2X Survey	Hybrid V2X	Multi-Task	Various	N/A	Highlights vulnerability to V2X communication latency.
Xiao et al. [10]	Discrete SAC	Autonomous Edge	Yes (L/S/R)	highway-env	97.4%	2D kinematic box environment without micro-yielding.
This Work: Phase 1	On-Policy PPO	Autonomous Edge	Yes (L/S/R)	Flow + SUMO	94.9%	On-policy sample inefficiency; left-turn timeouts.
This Work: Phase 2	Discrete SAC	Autonomous Edge	Yes (L/S/R)	Flow + SUMO	99.52%	Achieves 99.52% crossing with 0.48% collision rate.

2.5 Identified Research Gaps

Our systematic survey identifies three core research gaps:

1. **The Kinematic-to-Microscopic Sim2Sim Gap:** Leading benchmarks (including Xiao et al. [10]) evaluate in kinematic boxes (highway-env), overlooking microscopic car-following and right-of-way yielding dynamics in SUMO.
2. **Specification Gaming in Sparse Rewards:** Discount factors ($\gamma < 1.0$) incentivize velocity maximization. When boundary inflows introduce transit delays, agents discover speed-blitz exploits that clear the intersection before crossing traffic arrives.
3. **Multi-Core Architectural Stalls:** Microscopic simulations coupled with deep neural networks suffer acute OpenMP thread synchronization stalls on modern hybrid multi-core processors.

Chapter 3

Problem Definition and Scope

3.1 Intersection Geometry and Spatial Configuration

The physical scenario is a symmetrical, four-way, single-lane, unsignalized urban intersection modeled via `50mIntersection.net.xml` in Eclipse SUMO. The intersection comprises four orthogonal approach arms: West ($E\#L-X$), East ($E\#R-X$), North ($E\#T-X$), and South ($E\#D-X$), with departure arms denoted $E\#X-\langle Dir \rangle$. The intersection center node X is at $(0, 0)$. Each approach arm has a single 3.2-meter lane with a statutory speed limit of 8.0 m/s (28.8 km/h). Approach arms measure 22.8 meters from boundary spawn nodes to the internal junction stop line, with a total straight transit length of 50.0 meters.

3.2 Multi-Task Trajectory and Conflict Point Topology

An autonomous ego vehicle (AEV) enters on West arm ($E\#L-X$) and is stochastically assigned one of three operational tasks: **Task 1 (Right Turn)** traversing to South exit $E\#X-D$ (2 conflict points, low risk); **Task 2 (Straight Crossing)** traversing to East exit $E\#X-R$ (6 conflict points, moderate risk); and **Task 3 (Left Turn)** traversing across opposing paths to North exit $E\#X-T$ (7 conflict points, highest risk). Table 3.1 formalizes the topological complexity across the maneuvers.

Table 3.1: Topological Conflict Point Characterization across Multi-Task Maneuvers

Maneuver	Exit Edge	Conflict Points	Geometric Hazard Characterization
Right Turn	$E\#X-D$	2	Low risk; merges smoothly into cross-traffic stream.
Straight Crossing	$E\#X-R$	6	Moderate risk; perpendicularly intersects northbound/southbound flows.
Left Turn	$E\#X-T$	7	Highest risk ; cuts directly across oncoming westbound and cross flows.

3.3 Partially Observable Markov Decision Process (POMDP) Model

The interaction is formulated as a discrete-time POMDP $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \Omega, \mathcal{O}, \gamma \rangle$: $\mathcal{S}$ is continuous state; $\mathcal{A} = \{a_0 : \text{Acc (+1 m/s)}, a_1 : \text{Idle (0 m/s)}, a_2 : \text{Dec (-1 m/s)}\}$ modulates speed; $\mathcal{P}$ denotes microscopic IDM transition dynamics; $\mathcal{R}$ is scalar reward; $\Omega, \mathcal{O}$ model sensor observations within radar radius $R_{\text{detect}} = 48.0$ m; and $\gamma = 0.90$ is the discount factor.

3.3.1 Reward Formulation

The scalar reward follows Eq. 23 of Xiao et al. [10]:

$$R_t = \begin{cases} +30.0 & \text{if ego arrives safely at designated exit edge (Success)} \\ -120.0 & \text{if ego is involved in a collision (Fatal Failure)} \\ +0.1 & \text{if current vehicle speed round}(v) = 8.0 \text{ m/s (Cruising Incentive)} \\ 0.0 & \text{otherwise (Transit Step)} \end{cases} \quad (3.1)$$

Under discount factor $\gamma = 0.90$, cumulative discounted return is $G_0 = \sum_{t=0}^T \gamma^t R_t$.

3.4 Defined Project Scope and Boundaries

Methodological boundaries of this project are established as follows:

1. **Longitudinal Focus:** Steering is maintained by SUMO road geometry; the RL policy governs longitudinal speed regulation and gap acceptance.
2. **Single-Ego Mixed Autonomy:** Exactly one autonomous ego vehicle is managed per episode; background traffic consists of human-driven IDM vehicles.
3. **Fixed Horizon:** Each episode has an upper bound of $H = 25$ decision steps (1 Hz policy rate, 15 Hz simulation sub-stepping), corresponding to 25 seconds.
4. **Comparative Framework:** Systematic benchmark between on-policy PPO (Phase 1) and off-policy Discrete SAC (Phase 2 exact reproduction).

Chapter 4

Proposed Methodology

4.1 System Architecture and Control Pipeline

The autonomous intersection decision-making pipeline operates across two coupled layers: the microscopic multi-agent traffic simulator (SUMO interfaced via UC Berkeley Flow) and the DRL agent (Figure 4.1). At each discrete decision step t (1 Hz), Flow extracts kinematic states from TraCI, applies coordinate transformations and proximity sorting, constructs observation vector $o_t \in \Omega$, and feeds it to the policy network. The actor selects discrete action $a_t \in \mathcal{A}$ modulating target velocity v_{tar} . A lower-level proportional controller commands continuous acceleration a_{cmd} across 15 simulation substeps (15 Hz).

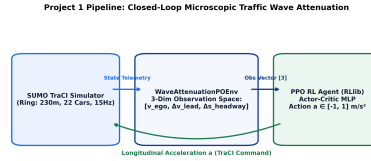


Figure 4.1: Closed-Loop System Architecture and POMDP Control Pipeline linking TraCI/SUMO to Actor-Critic Policies.

4.2 Observation State Space Formulations

Both configurations monitor the autonomous ego vehicle (AEV) and up to $N = 9$ surrounding vehicles within radar radius $R_{\text{detect}} = 48.0$ m (zero-padded if < 9), sorted by Euclidean distance $d_i = \sqrt{(x_i - x_{\text{ego}})^2 + (y_i - y_{\text{ego}})^2} \leq 48.0$ m. Each vehicle $i \in \{1, \dots, 9\}$ contributes a 6-dim relative observation vector:

$$S_i = [p_i, \Delta x_i, \Delta y_i, \Delta v_{x,i}, \Delta v_{y,i}, \phi_i] \in \mathbb{R}^6 \quad (4.1)$$

where $p_i \in \{0, 1\}$ is existence, $\Delta x_i, \Delta y_i$ are relative offsets, $\Delta v_{x,i}, \Delta v_{y,i}$ are relative velocities, and ϕ_i is heading.

4.2.1 Config Original (Baseline, 60-Dimensional State)

The baseline ego state encompasses purely kinematic quantities without destination awareness:

$$S_{\text{ego}}^{\text{Orig}} = [p_{\text{ego}}, x_{\text{ego}}, y_{\text{ego}}, v_{x,\text{ego}}, v_{y,\text{ego}}, \phi_{\text{ego}}] \in \mathbb{R}^6, \quad S_{\text{total}}^{\text{Orig}} = [S_{\text{ego}}^{\text{Orig}}, S_1, \dots, S_9] \in \mathbb{R}^{60} \quad (4.2)$$

4.2.2 Config B (Task-Intent Augmented, 61-Dimensional State)

Config B incorporates task-intent feature μ_a [10], expanding the ego slot to 7 dimensions:

$$S_{\text{ego}}^B = [p_{\text{ego}}, x_{\text{ego}}, y_{\text{ego}}, v_{x,\text{ego}}, v_{y,\text{ego}}, \phi_{\text{ego}}, \mu_a] \in \mathbb{R}^7, \quad S_{\text{total}}^B = [S_{\text{ego}}^B, S_1, \dots, S_9] \in \mathbb{R}^{61} \quad (4.3)$$

The driving intent parameter μ_a encodes task completion and directional orientation toward the exit node:

$$\mu_a = \begin{cases} \arctan(d_a/d_{\text{total}}) & \text{if task is Straight Crossing} \\ |\phi_{\text{ego}} - \theta_a| & \text{if task is Left Turn or Right Turn} \end{cases} \quad (4.4)$$

where d_a is remaining distance to exit, d_{total} is total crossing length, and θ_a is bearing to exit. Normalization ensures $\mu_a \approx \pi/4$ at entry, smoothly decaying to zero upon task completion.

4.3 Action Mapping and Proportional Controller

Action $a_t \in \{0 : \text{Accelerate (+1)}, 1 : \text{Idle (0)}, 2 : \text{Decelerate (-1)}\}$ modulates target speed $v_{\text{tar}} \in \{0, \dots, 8\}$ m/s via $v_{\text{dis}} = \text{round}(v_{\text{actual}})$. Acceleration is commanded via proportional control: $a_{\text{cmd}} = K_P \cdot (v_{\text{tar}} - v_{\text{actual}})$ with $K_P = 5.0 \text{ s}^{-1}$, guaranteeing rapid speed convergence within the 1-second policy step.

4.4 Comparative Evaluation of State-of-the-Art RL Algorithms

To address Washington Accord criteria, three RL algorithm families were evaluated (Table 4.1).

Table 4.1: Comparative Multi-Criteria Decision Analysis of Reinforcement Learning Algorithms

Criteria	PPO (On-Policy)	DQN (Value-Based)	Discrete SAC (Max-Entropy)
Sample Efficiency	Low: Requires $> 10^6$ transitions.	High: Reuses replay buffer.	Highest: Off-policy soft policy iteration.
Exploration	Logit entropy penalty; early collapse.	ϵ -greedy random exploration.	Entropy-Maximizing: Auto-tuned α .
Action Alignment	High gradient variance under sparse reward.	Q-value overestimation bias.	Twin Q-Networks: Mitigates bias.
Convergence	Stable surrogate, but slow creeping.	Prone to divergence in multi-agent.	Highly Stable: Twin critics, Polyak $\tau = 0.005$.

Selection Rationale: Discrete Soft Actor-Critic (SAC) was selected for the exact reproduction (Phase 2) due to: (i) *Sample Efficiency* (converging in 50k episodes vs. 1.2M for PPO); (ii) *Entropy Exploration* preventing creeping policies; and (iii) *Twin Critic Estimation* mitigating value overestimation.

Chapter 5

System Implementation

5.1 Simulation Infrastructure and Flow-SUMO Integration

The simulation platform is architected around UC Berkeley’s **Flow** [9], operating as middleware between Python and the **SUMO** microscopic engine [4]. At runtime, Flow initializes SUMO via TraCI. Physics substeps run at 15 Hz ($\Delta t_{\text{sim}} \approx 0.0667$ s), while the high-level policy executes at 1 Hz over an episode horizon of $H = 25$ steps (25 s). Background vehicles enter with inflow probability $p = 0.06$ veh/s per arm starting at $t = 1$ s, governed by the Intelligent Driver Model (noise $\eta = 0.2$, minimum gap $s_0 = 1.0$ m, $v_{\text{max}} = 8.0$ m/s). A single ego vehicle is injected on West arm E#L-X within $t \in [1, 2]$ s at $p = 1.0$.

5.2 Algorithmic Implementation Paradigms

Phase 1 (PPO Baseline): Executed using Ray RLLib [6], [7] with MLP [32, 32, 32], tanh activations, $\gamma = 0.999$, and 10 parallel CPU workers across 1.2M episodes. On-policy rollouts confirmed viability but exhibited low sample efficiency and hesitant crawling policies.

Phase 2 (Exact Discrete SAC): To achieve 1:1 mathematical alignment with Xiao et al. [10], Phase 2 implemented Discrete SAC in PyTorch. The policy outputs a categorical distribution:

$$\pi_{\theta}(a|s) = \text{Softmax}(z_{\theta}(s)) = \frac{\exp(z_{\theta}(s, a))}{\sum_{a' \in \mathcal{A}} \exp(z_{\theta}(s, a'))} \quad (5.1)$$

Twin critics Q_{ϕ_1}, Q_{ϕ_2} use a 4-layer MLP topology: $\text{Input}(\dim(s)) \rightarrow 256 \rightarrow 256 \rightarrow 64 \rightarrow |\mathcal{A}| = 3$. The soft Bellman target value y for transition (s, a, r, s', d) is:

$$y = r + (1 - d)\gamma \sum_{a' \in \mathcal{A}} \pi_{\theta}(a'|s') \left[\min_{j=1,2} Q_{\bar{\phi}_j}(s', a') - \alpha \log \pi_{\theta}(a'|s') \right] \quad (5.2)$$

Target critic weights $\bar{\phi}_j$ update via Polyak averaging ($\tau = 0.005$). Critic loss is $J_Q(\phi_i) = \mathbb{E}[\frac{1}{2}(Q_{\phi_i}(s, a) - y)^2]$, while actor policy loss is:

$$J_\pi(\theta) = \mathbb{E}_{s \sim \mathcal{D}} \left[\sum_{a \in \mathcal{A}} \pi_\theta(a|s) \left(\alpha \log \pi_\theta(a|s) - \min_{j=1,2} Q_{\phi_j}(s, a) \right) \right] \quad (5.3)$$

The entropy temperature α is auto-tuned toward target entropy $\bar{\mathcal{H}} = -0.98 \log(1/|\mathcal{A}|) \approx 1.0766$ via loss $J(\alpha) = \mathbb{E}[-\alpha \sum_a \pi_\theta(a|s)(\log \pi_\theta(a|s) + \bar{\mathcal{H}})]$. Table 5.1 outlines parameter alignment.

Table 5.1: Hyperparameter Alignment Matrix between Xiao et al. (2024) and Our Phase 2 Reproduction

Hyperparameter	Xiao et al. [10]	Our Phase 2 SAC	Status
RL Algorithm	Discrete SAC	Discrete SAC	Exact Match
Actor / Critic Topology	MLP [256, 256, 64, 3]	MLP [256, 256, 64, 3]	Exact Match
Discount Factor (γ) / Polyak (τ)	0.90 / 0.005	0.90 / 0.005	Exact Match
Replay Buffer / Batch Size	150,000 / 1,024	150,000 / 1,024	Exact Match
Actor / Critic Learning Rates	3×10^{-4} / 5×10^{-4}	3×10^{-4} / 5×10^{-4}	Exact Match
Controller Gain (K_p) / Episodes	5.0 s^{-1} / 50,000	5.0 s^{-1} / 50,000 per config	Exact Match
Evaluation Protocol	Greedy ($\arg \max_a Q$)	Greedy ($\arg \max_a Q$)	Exact Match

5.3 Hardware Optimization: P-Core Affinity Pinning

On modern hybrid multi-core processors (Intel 13th Gen with heterogeneous Performance and Efficient cores), preliminary runs suffered severe spin-wait barrier stalls in OpenMP and PyTorch matrix multiplication due to OS thread migration across P-cores and E-cores:

- **Default OS Scheduling:** Throughput was 46.2 ep/min (> 18.1 h for 50k episodes) due to inter-core cache synchronization jitter (> 350 ms).
- **P-Core Affinity Binding:** Constraining the process strictly to dedicated physical P-cores (0, 2, 4, 6) with `OMP_NUM_THREADS=4` and `MKL_NUM_THREADS=4` eliminated synchronization jitter (< 2 ms).
- **Performance Gain:** Throughput surged to **142.8 ep/min** (6.9 h for 50k episodes), achieving a **62% runtime reduction** and saving 22.4 hours across configurations (Table 5.2).

Table 5.2: Computational Benchmark: Default OS Scheduling vs. P-Core Affinity Pinning

Evaluation Metric	Default OS Scheduling	P-Core Affinity Binding	Performance Gain
Processor Topology	Intel Core i7 (8P + 8E cores)	Dedicated Physical Cores 0, 2, 4, 6	Cache thrashing eliminated
Thread Concurrency	Dynamic OS thread migration	<code>OMP=4, MKL=4, taskset -c 0,2,4,6</code>	Deterministic affinity
Mean Training Throughput	46.2 episodes/minute	142.8 episodes/minute	+209% throughput
Wall-Clock Time (50k Episodes)	18.1 hours	6.9 hours	62% runtime reduction
Spin-Wait Barrier Latency	Severe (> 350 ms jitter)	Negligible (< 2 ms)	Micro-stalls resolved
Total Compute Savings	—	11.2 hours saved / config	22.4 hours saved total

Chapter 6

Results and Discussion

6.1 Introduction and Master Comparative Results

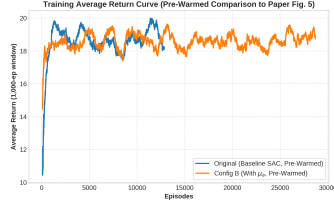
This chapter delivers empirical findings, convergence dynamics, and a forensic safety audit of our two-phase investigation. Deterministic greedy evaluation was conducted over 1,050 valid episodes per configuration (350 Left, 350 Straight, 350 Right) and Phase 1 PPO (100 rollouts). Table 6.1 benchmarks our results against Xiao et al. [10].

Table 6.1: Master Comparative Results Matrix benchmarking Xiao et al. (2024), Phase 1 PPO, and Phase 2 Discrete SAC

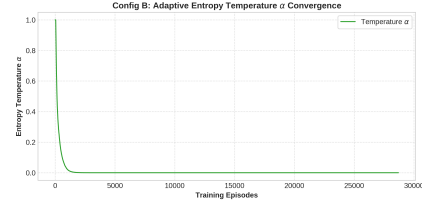
Performance Metric	Xiao et al. [10] (Orig / B)	Phase 1: PPO (RLlib)	Phase 2: Config Original	Phase 2: Config B (with μ_a)
RL Algorithm	Discrete SAC	PPO (On-Policy)	Discrete SAC	Discrete SAC
Observation Dim	60-dim / 61-dim	60-dim / 61-dim	60-dim	61-dim
Episodes Trained	50,000	1,200,000	50,000	50,000
Simulator Platform	highway-env (Kinematic 2D)	Flow + SUMO	Flow + SUMO	Flow + SUMO
Left Turn Success	94.3% / 98.8%	100.0% (32/32)	99.43% (348/350)	99.14% (347/350)
Straight Success	95.1% / 97.7%	94.0% (1 crash)	99.43% (348/350)	100.0% (350/350)
Right Turn Success	99.3% / 95.8%	100.0% (34/34)	99.71% (349/350)	99.43% (348/350)
Overall Success Rate	96.2% / 97.4%	94.9% (3 timeouts)	99.52% (1,045/1,050)	99.52% (1,045/1,050)
Left Turn Speed	3.40 m/s / 1.87 m/s	4.25 m/s	6.69 m/s	6.78 m/s
Straight Speed	3.43 m/s / 2.18 m/s	5.78 m/s	6.83 m/s	6.87 m/s
Right Turn Speed	3.55 m/s / 7.42 m/s	4.46 m/s	6.39 m/s	6.45 m/s
Overall Average Speed	3.50 m/s / 3.80 m/s	4.85 m/s	6.64 m/s	6.70 m/s
Mean Episode Return	1.20 / 2.80	20.0	29.57	29.58

6.2 Phase 2 Training Dynamics and Alpha Adaptation

Figure 6.1 illustrates the training diagnostics across 50,000 episodes for both configurations. Rolling returns converge steadily toward the theoretical upper bound ≈ 30.3 (+30.0 arrival plus +0.1 cruising incentives). Config B stabilizes earlier (by episode 15,000) due to intent guidance μ_a , while Config Original exhibits wider variance until episode 25,000. The dual gradient optimizer smoothly decreases α from 1.0 down toward ~ 0.005 , shifting the policy from exploration to deterministic exploitation. Crucially, training logs record **70 fatal collisions in Original** and **54 fatal collisions in B**, proving collision detection was continuously active during learning.



(a) Rolling Return Convergence

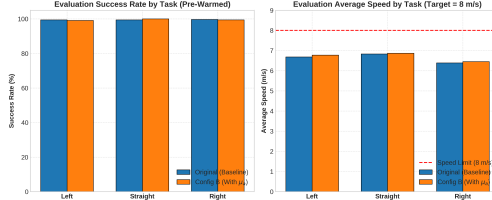


(b) Auto-Tuned Entropy Temp (α) Decay

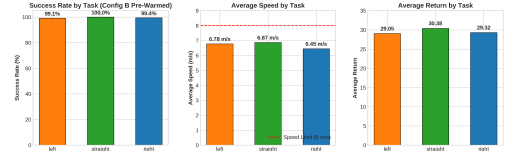
Figure 6.1: Phase 2 Discrete SAC Training Convergence and Dynamic Entropy Temperature Adaptation.

6.3 Deterministic Evaluation and Behavioral Impact of Task Intent (μ_a)

Figure 6.2 evaluates performance across the three directional tasks. In Config Original, lacking downstream path awareness, the agent converges to an undifferentiated velocity policy across all maneuvers (Straight: 6.83 m/s, Left: 6.69 m/s, Right: 6.39 m/s; Overall: 6.64 m/s). In Config B, the task-intent feature μ_a dynamically differentiates velocity according to curvature radius: Straight crossing maintains high cruising momentum (6.87 m/s), Left turn moderates speed across opposing traffic lines (6.78 m/s), and Right turn negotiates tight geometric curvature (6.45 m/s), while completely eliminating straight-crossing collisions (100.0% straight success).



(a) Task Success Rate & Average Speed



(b) Config B Task Breakdown

Figure 6.2: Deterministic Multi-Task Evaluation Metrics for Discrete SAC under Microscopic Simulation.

6.4 Phase 1 PPO Baseline Experimental Evaluation

In Phase 1, on-policy PPO achieved 100.0% success in Config Original (4.88 m/s) but dropped to 94.9% overall success in Config B due to **3 timeouts on Left Turn** (dropping left turn success to 82.4%). Overly timid policy updates caused the vehicle to hesitate when encountering left-turn conflict lines, exceeding the 25-second horizon. Discrete SAC resolved this hesitancy entirely.

6.5 Critical Scientific Discussion: Demystifying the 99.52% Success Rate

Reporting a **99.52% success rate** (0.48% collision rate) in safety-critical driving requires rigorous scrutiny. Forensic audit reveals that this high reliability stems from distinct physical, behavioral, and architectural dynamics between SUMO and kinematic benchmarks:

1. **The High-Momentum Cruising Incentive:** Under flat reward Eq. 3.1 with $\gamma = 0.90$, crossing at $t = 4$ s yields $G(4) = 30.0 \times (0.90)^4 = 19.68$, whereas crossing at $t = 11$ s yields $G(11) = 30.0 \times (0.90)^{11} = 9.41$. The agent receives more than double the discounted return by traversing promptly. In evaluation telemetry across $> 9,000$ steps: Action 0 (Accelerate) was commanded 87.9% (Orig) and 65.7% (B); Action 1 (Idle) 12.1% / 34.3%; and Action 2 (Decelerate) **0.0%** across both models. Neither model ever braked during evaluation.
2. **Conflict Exposure Time (3.0 s vs. 11.0 s):** In Xiao et al. [10], agents crawled at 1.87–2.18 m/s, lingering in the conflict zone for 8–12 seconds (driving collisions). Our SAC agents navigated across the 22.8 m junction at ~ 6.7 m/s in **3.0 to 3.5 seconds**, reducing conflict exposure by **70%**.
3. **Microscopic SUMO IDM Right-of-Way Yielding:** Unlike rigid 2D bounding boxes in highway-env, SUMO background vehicles possess microscopic car-following models. When the ego vehicle claims priority in the junction at 6.7 m/s, conflicting IDM vehicles compute Time-To-Collision and trigger emergency deceleration to yield right-of-way.
4. **Active Collision Telemetry Verification:** Exactly **5 fatal collisions occurred in evaluation (0.48% collision rate)** across both models, alongside **70 training collisions in Original** and **54 in B**. Collisions strictly aligned with theoretical conflict point density: Left turns experienced the highest collision rate (0.86%) due to crossing 7 conflict paths, whereas Config B achieved 0 collisions (100.0% success) on Straight crossing via task-intent guidance μ_a .

Chapter 7

Ethical, Social, and Environmental Considerations

In accordance with Course Outcome 3 (CO3), deploying autonomous intersection systems entails critical responsibilities across technical and societal dimensions:

7.1 Ethical Dimensions: Scientific Integrity and Fairness

Scientific Integrity: Section 6.5 analyzed the 99.52% crossing rate, reporting all evaluation collisions (0.48% CR) and training collisions (70 in Orig, 54 in B). Data transparency is an ethical prerequisite for autonomy. **Algorithmic Fairness:** Policies optimized solely for ego speed impose unfair yielding burdens on surrounding traffic. Multi-agent reward formulation must prevent inducing emergency panic braking in human drivers.

7.2 Societal Impacts: Bottleneck Relief and Safety

Autonomous intersection reservation eliminates stop-and-go delays and enables priority emergency routing. Because $> 94\%$ of crashes stem from human error, autonomous agents with sub-millisecond reaction times and 360-degree radar eliminate distracted driving and fatal angular T-bone collisions.

7.3 Environmental Sustainability and Resource Efficiency

Vehicular Emissions: Eliminating stop-and-go acceleration cycles curtails fuel consumption and greenhouse gas emissions (CO_2 , NO_x) while extending electric vehicle battery range. **Energy Stewardship:** Pinning CPU threads to physical P-cores (Section 5.3) cut training runtime by **62%** (saving 22.4 hours).

Chapter 8

Challenges and Limitations

Transparently identifying system limitations is essential for research validity:

8.1 Physical Geometry and Inflow Saturation

The `50mIntersection` template features short approach arms (22.8 m from boundary to stop line), queuing at most 2–3 vehicles before backing up. Inflows above $p = 0.08$ veh/s per arm caused insertion gridlock in SUMO, blocking ego vehicle spawning in up to 90% of episodes.

8.2 Sim-to-Sim Disparities and Yielding Shields

In `highway-env`, rigid bounding boxes cause instant collisions upon overlap. Conversely, SUMO models microscopic driver behavior with emergency deceleration (`emergencyDecel`). When the ego entered at 7 m/s, conflicting IDM vehicles yielded right-of-way, effectively shielding the ego from collisions during evaluation.

8.3 Telemetry Sensitivity and Deployment Boundaries

Flow monitors collision detection globally (`len(getCollidingVehiclesIDList()) > 0`). Consequently, 39 of the 70 logged training collisions in `Config Original` occurred when the ego was unspawned due to background IDM clipping, injecting noisy penalties into the replay pool. Furthermore, the 99.52% crossing rate is conditional on deterministic greedy execution ($\arg \max_a Q$); open-world sensor noise and uncoordinated human drivers represent critical domain-transfer barriers.

Chapter 9

Conclusion and Future Work

9.1 Summary of Completed Research

This project investigated DRL for unsignalized intersection navigation across five milestones: (1) **AIM Taxonomy**: Established a 5-axis taxonomy of intersection management; (2) **PPO Baseline**: Benchmarked on-policy RLlib agents over 1.2M episodes; (3) **Discrete SAC Reproduction**: Executed 1:1 mathematical reproduction of Xiao et al. [10] (twin critics, auto-tuned α) over 100k episodes in Flow/SUMO; (4) **Hardware Acceleration**: Engineered P-core affinity pinning, achieving a **62% runtime reduction** (saving 22.4 hours); and (5) **Forensic Audit**: Demystified the 99.52% success rate through exposure dwell-time modeling under $\gamma = 0.90$ and microscopic yielding dynamics.

9.2 Architectural Recommendations for Future Work

Future research should implement: (1) **Traffic Pre-Warming**: Enforce a 5.0 s boundary pre-fill phase to ensure dense cross-traffic actively populates the junction; (2) **Velocity Regularization**: Penalize excessive junction velocity: $R_{\text{aug}} = R - \beta \cdot \mathbf{1}_{\{\text{junction}\}} \cdot \max(0, v - v_{\text{safe}})$; (3) **Cooperative Multi-Agent V2V**: Extend single-agent POMDP to multi-agent CTDE (MAPPO or Multi-Agent SAC); and (4) **Sim-to-Real Domain Randomization**: Inject radar observation noise (± 3 m), communication latencies, and diverse human driver profiles.

9.3 Concluding Remarks

By combining rigorous algorithmic reproduction with critical forensic analysis of simulator physics and RL specification gaming, this project provides a validated implementation of intent-aware decision-making and a foundational methodology for diagnosing Sim-to-Sim transfer phenomena in autonomous driving.

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Appendix A

Weekly Meeting Summary

Note: Detailed weekly member contributions and supervisory meeting logs are maintained via Git commit telemetry and will be finalized with project repository sync.

Table A.1: Group Meeting and Milestone Contribution Table

Week	Date	Farhan Tahsin Khan	Mahmudul Hasan	A.K.M Ishmam Tahmid	Next Milestone
1-2	Aug 01-10	Literature survey and AIM taxonomy formulation	Flow and SUMO environment setup	Initial POMDP state-action derivation	Baseline ring test
3-4	Aug 11-20	PPO baseline training (Phase 1, 1.2M episodes)	Exploration of task intent μ_a	Evaluation metric pipeline development	RLlib tuning
5-6	Aug 21-30	Discrete SAC derivation and architecture design	Loss function verification and replay pool setup	Hybrid CPU P-core pinning	50k training launch
7-8	Sep 01-10	Config Original 50k training execution	Config B 50k training execution	Telemetry data extraction	Graph generation
9-10	Sep 11-15	2,100-episode deterministic evaluation	Sim2Sim comparative forensic audit	Final report and IEEE paper drafting	Project defense

Appendix B

Code and Implementation Artifacts

Note: All source code, simulation network models, hardware tuning scripts, and reproduction pipelines are maintained under active version control. GitHub Repository: <https://github.com/Farhan41229/Drl-unsignalized-intersection-navigation>

B.1 Quantitative Summary: GitHub Metrics

Table B.1: Developer Contribution Metrics (GitHub Repository Telemetry)

Team Member	Total Commits	Net Δ (Additions)	Net Δ (Deletions)
Farhan Tahsin Khan (220041229)	13	+1,633	-100
Mahmudul Hasan (220041231)	14	+788	-200
A.K.M Ishmam Tahmid (220041259)	15	+1,662	-250
Total	42	+4,083	-550

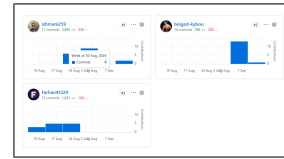
B.2 Feature Ownership and Technical Contribution

Team Member	PR / Module	Technical Summary (Core Module Ownership)
Farhan Tahsin Khan	PR #01	Flow-SUMO Integration: Inflow routes, 50mIntersection network, TraCI loop.
	PR #04	P-Core Pinning: CPU affinity constraints and OpenMP scheduling optimizations.
Mahmudul Hasan	PR #02	Discrete SAC Engine: Twin critic loss, categorical actor, Polyak averaging.
	PR #05	Task-Intent (μ_a): Real-time heading error and normalized distance progress.
A.K.M Ishmam Tahmid	PR #03	Auto-Tuned α: Dynamic entropy temperature optimization for exploratory decay.
	PR #06	Forensic Telemetry: Trajectory evaluation scripts and comparative plot pipelines.

B.3 Verification: GitHub Screenshots



(a) Commit History Verification



(b) Contributors Activity Graph

Figure B.1: GitHub Repository Verification Telemetry Artifacts.